Project Experience Report

SSE Capstone: University of Regina

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ClassQuest

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2.0 Introduction

2.1 Problem

What was the problem to solve and what was our solution?

Classrooms today face a lot of challenges in maintaining student engagement. Without proper tools to support attention, motivation and participation, teachers often struggle to keep students actively involved in class activities. The current platforms focus on administrative tasks instead of maintaining an engaging experience for students. This is where we identified a gap, teachers need a way to make learning more interactive and rewarding, while students need to benefit from a system that recognizes hard work, encourages collaboration and makes progress visible. Our solution "*ClassQuest"* addresses this gap, by introducing an exciting user interface that transforms everyday classroom tasks into a rewarding RPG game-like experience.

2.2 Target users and Stakeholders

Who were the users, who was involved?

Ideally, this project should target all schools. However, due to time constraints and advice from our project facilitator Prof. Tim Maciag, the scope was narrowed down to Saskatchewan classrooms targeting teachers and students from grades 5 to 8. The main stakeholders for this project include Alvin Tolentino, Tolani Oke-Steve, Dmytro Stepaniuk, our advisors (public school teachers) Brian Nenson and Kelly Chambers. Other stakeholders include Prof. Timothy Maciag our capstone facilitator and Prof. Yogesh Sharma our project mentor.

2.3 Scope and Objectives

What were the scope and objectives?

The scope of our project was defined to ensure that we could deliver a fully functioning system within our limited timeline. From the start, rather than attempting

to duplicate the complexity of existing educational software programs, we reduced our scope to building a refined system that managed a clear gap, the lack of tools designed to support student motivation, active participation and visual feedback on progress. This meant placing priority on features that enhance the educational experience for both students and teachers while keeping the core implementation achievable.

Our major objective during this project was to design a platform that helps teachers make learning more interactive while maintaining an engaging learning experience for students. To achieve this, we integrated specific gamification elements such as experience points (XP), gold, levels and a stats manager that encourage consistent participation while preserving the core purpose of learning. For teachers the goal was to create an easy to navigate interface that made managing classroom activities more efficient.

On the technical side, our major objective was to deliver a scalable full stack application with essential components such as authentication, data storage, and responsive UI design, while intentionally excluding advanced features that could not be completed within the timeframe.

3.0 Outcome

What actually occurred?

While we successfully implemented the core features of the platform, several components, particularly the boss battle system required significantly more time and multiple redesigns than initially anticipated due to technical and logical complexities.

3.1 Problem

Evolution of our problem statement

Throughout the duration of this project, the problem statement for our project **ClassQuest** stayed the same, classrooms lacked tools that motivate students and support teachers in managing participation, and our core purpose was to address this gap by developing an RPG gamified learning platform that makes classroom activities more fun.

While our core problem did not change, our understanding of solving it changed greatly. At the start of the project, we visualized an extensive platform with features for students, teachers and parents. However, based on feedback from our stakeholders and timing constraints we simplified our solution to deliver the most impactful elements some of which included (quests, boss battles, experience points and a character customization model). These elements became the base of the ClassQuest model. On the implementation side, our solution developed from a concept to a fully working use case implemented using TypeScript and AWS. We successfully built a full-stack application that allowed teachers to create, manage and assign boss battles and quests to students who received rewards upon completion of assigned tasks.

3.2 Target users and Stakeholders

Evolution of target users and stakeholders

Same as above, our target audience and stakeholders remained consistent. From the start, we focused on Saskatchewan classrooms specifically teachers and students from grades 5 to 8, this narrowed scope allowed us to customize our application to a realistic and attainable user base within the given time frame. While our long term vision still included delivering our platform to more schools, we kept our implementation strategy focused on our target users to ensure proper quality and usability. This consistency in target users and stakeholders allowed us to maintain a

good structure between our design decision, implementation goals and our final deliverables.

3.3 Scope and Objectives

How did the scope & objectives evolve?

Our core objectives remained unchanged, but our scope evolved in response to constraints encountered. From the start, our core objective was building a gamified learning platform that enhances student motivation and participation in the classroom. This understanding helped us to stay aligned with our original version and avoid scope creep. With this, our goals became clearer and our visions for what ClassQuest should be became more unified. Initially, we considered extensive features such as the parent side of the application and a stats manager section for students. However, due to timing constraints and feedback, we reduced our scope to include the core gamification elements such as quests, boss battles, experience points, gold and hearts. These features were chosen for their direct link to engagement and their achievability within our timeframe. This ensured that our objectives remained achievable and implementation was scalable for future developments. Technical challenges also influenced our scope. For example, the boss battle system required more structures than initially planned to include a lobby feature, guild organization, and real-time synchronization between students and teachers. We underestimated the implementation of these features especially integrating AppSync and GraphQL for real-time updates. This resulted in multiple changes in the system to ensure proper functionality and user experience.

4.0 System Results

Product produced

4.1 Main page

What was the progress?

The main ClassQuest page serves as the entry point for both students and teachers. Initially our design for this page was too formal and lacked any gamification element, but with constant feedback we've redesigned the home page into a dynamic landing page with a responsive design that reflects the RPG theme. The final version incorporates gamification elements such as character previews and essential onboarding information to create an immersive first impression.

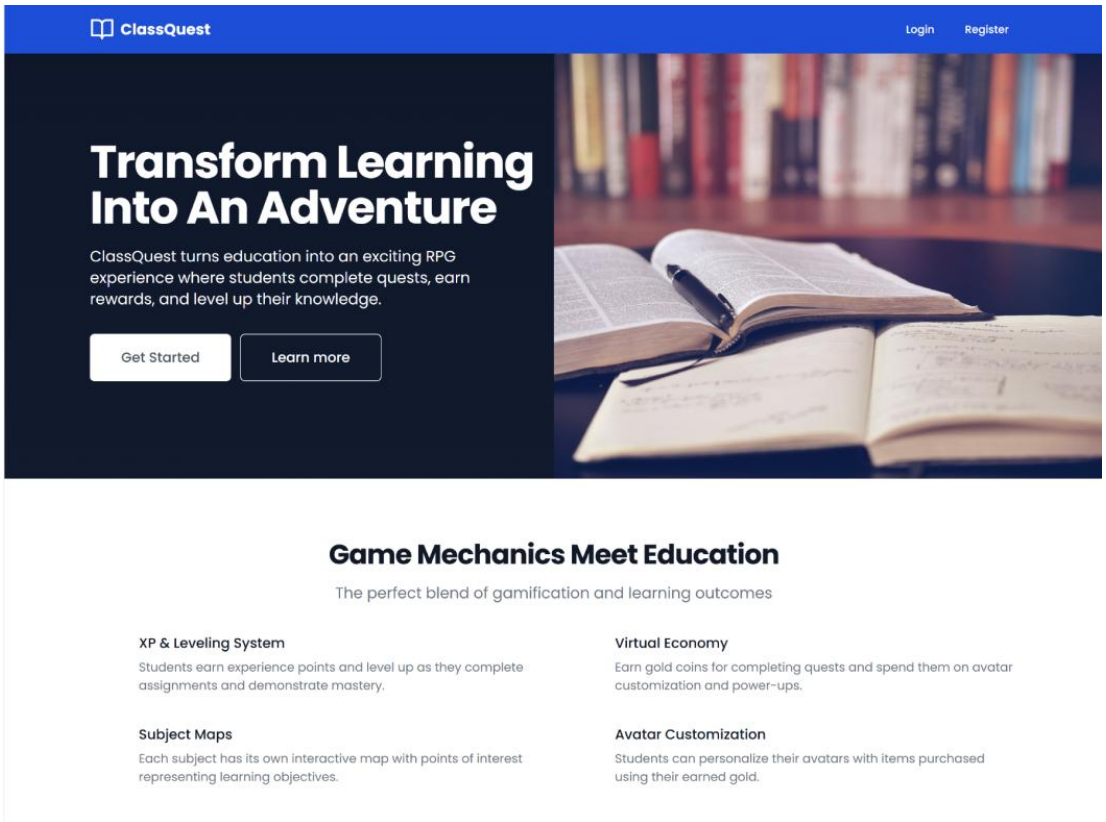

Figure 1: initial landing page

Figure 2: Updated landing page

4.2 Teacher Dashboard

The teacher dashboard page was one of the most essential pages to implement because it allows teachers to monitor active students, classes and quests, while also viewing recent activity across all classrooms and managing top students. This dashboard displayed below is the backbone for classroom management within ClassQuest.

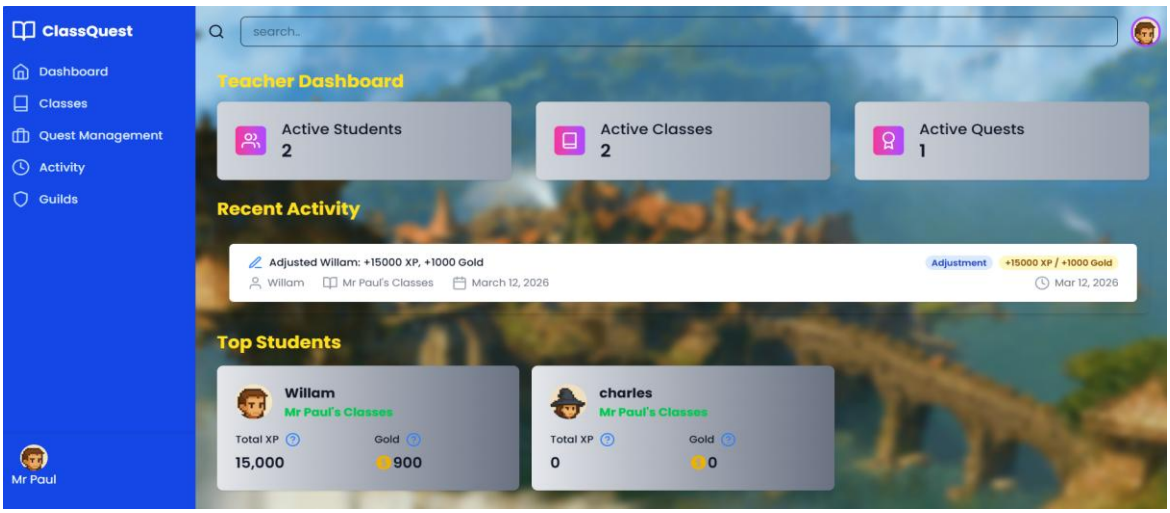

Figure 3: Teacher Dashboard

4.3 Quest Creation flow

Quest creation was a central feature from the beginning. Our prototype did not have the idea of this fully fleshed out, but during the implementation phase we streamlined this section to make it loosely coupled from classes and added backend support for proper storage. The final version allows teachers to create engaging quests and boss battles that are seamlessly integrated with the student's experience.

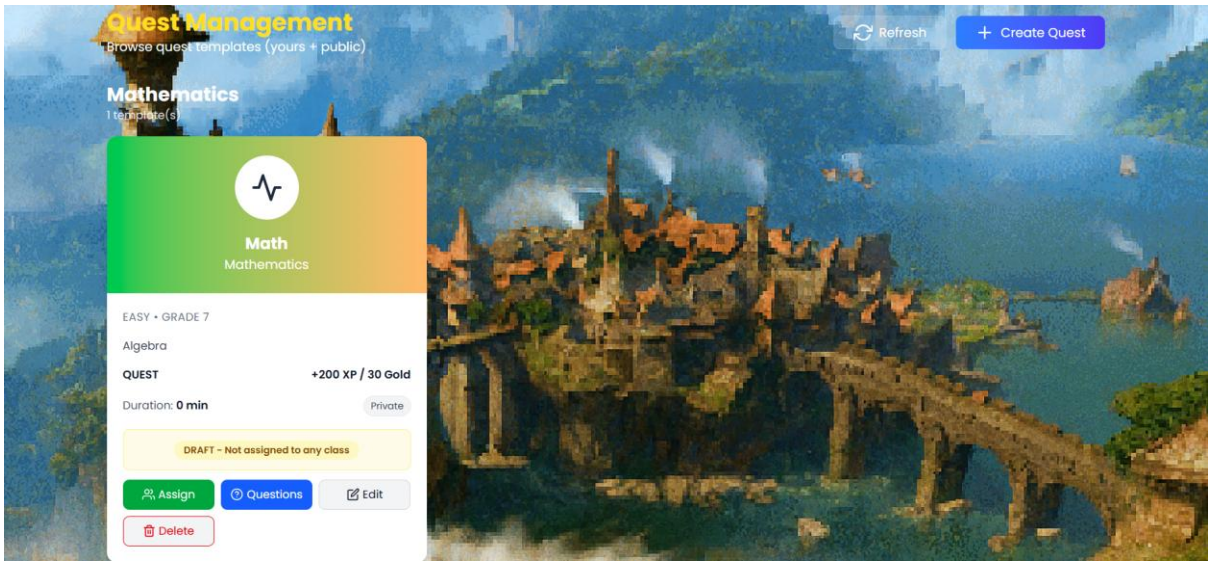

Figure 4: Quest Management

4.4 Guilds and Boss battles

Guilds and boss battles were designed with the intention of fostering collaboration in the classroom. Our early implementations were focused on visual immersion, but as we built out the system we shifted to more of an informative UI that still had elements of visual immersion as shown below. This page was tested with selected users to ensure that the desired intention was achieved.

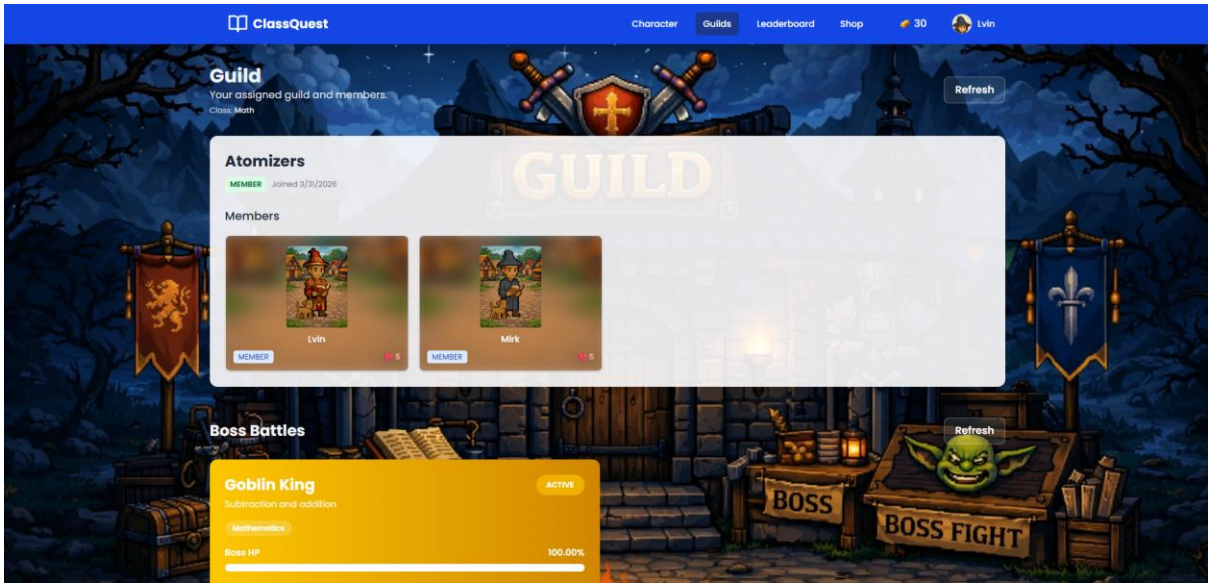

Figure 5: Guilds & boss battles

4.5 Technology Stack

ClassQuest is powered by a cloud native architecture built completely on AWS. Our system uses AppSync (GraphQL) as the backend API layer for flexible data querying, DynamoDB for extensible NoSQL storage, Lambda for the game logic, S3 buckets and CloudFront for image storage and retrievals, EventBridge for event-driven actions and Cognito for authentication. On the frontend we used React (TypeScript) running on Vite for fast development and optimized builds. We chose React for its component-based architecture and efficiency in creating dynamic user interfaces, TypeScript was used to add type safety. While Vite provided rapid reloading and minimal configuration. These divisions of responsibilities with React on the frontend and AWS on the backend ensured that the platform remained responsive and highly scalable throughout our development cycle.

5.0 Project Reflections

5.1 Overall Experience

What was our experience?

Working on this project was both a fun and rewarding experience, we got to see our idea turn into a real, interactive system that is fully functional. Building a full-stack application from the ground up under real world constraints like Cognito auth flows and DynamoDB modelling challenged us to think intently about the system architecture and scalability. We transitioned from static pages to a fully working product where quests, classes, guilds and the in-game economy all worked together effortlessly. This required careful coordination of frontend and backend for future growth. Beyond technical work, Alvin took on the challenge of making 2D characters with Krita, experimenting with styles that matched our RPG theme. These characters became a core element of the ClassQuest experience, and our meetings with Saurabh were essential in helping us understand Intellectual Property ownership and ethical use of assets.

Throughout development, we followed the agile workflow with weekly sprints and scrums to guide our priorities and design decisions. Working with our stakeholders-teachers forced us to shift our mindset from feature completion to product readiness, focusing on demo stability, user experience and proper documentation. Ultimately this project improved our understanding of system design trade-offs. Working on this project also improved our technical knowledge and gave us confidence in building user centered applications.

5.2 What went well

What were our achievements?

A good number of things went well for us, but one of the most rewarding aspects for us is that we never ran out of ideas, we always had new ways to improve our application whether through UI changes, fine tuning the backend or polishing

gameplay mechanics. We also gained more hands-on experience working with AWS, React and TypeScript. Plus, we explored AI tools to aid in development and design. These tools not only helped us make our system more scalable, but they also strengthened our understanding of this web architecture.

On the technical side, our core systems became functional, including quests, boss battles, classes, rewards, progression tracking and guilds. Our backend structure also became finalized after multiple refactorings into separate stacks, and we implemented features such as reward transactions, students states and tracking. We progressed from basic CRUD to operations to game-like mechanics such as XP, hearts and a progression system. This shift made responsibilities clearer and more improved our team alignment, with this frontend and backend integration became smoother reducing broken flows in our system and improving overall stability.

We also excelled in project management, meeting all our set deadlines consistently while mostly maintaining a status of green throughout our scrums. Communication was a big part of this project, and our Discord communication channel played a big role in keeping track of distributed tasks which contributed to us reaching MVP level completeness for all planned features before bazaar day. One standout achievement for us was the regular feedback from our stakeholders that validated our direction and confirmed the need for a gamification element in the classroom setting.

5.3 Our Challenges

What were the downs?

One of the most challenging features we had to implement was the boss battle system. While our initial idea was simple: students team up with their guilds to answer practice exam questions the actual implementation for this logic was far more complex than expected. This caused us to underestimate how many supporting pages we needed, including a lobby for students to join battle, guild-based categorization, and real-time synchronization between teachers and students during boss battles. Handling absent students in boss battles also added another layer of logic because this required adding rules for participation, penalties and rewards. Syncing these pages with AppSync and GraphQL was especially challenging since we didn't have any prior experience working with these tools.

Beyond the boss battle implementation, we also faced challenges with debugging because as our codebase grew, correcting issues and making small changes became more difficult. Integration between the frontend and the backend also caused recurrent issues, especially under AWS infrastructure limits like CloudFormation size and resource limits. Authentication and user flow design required multiple iterations to fix usability issues, the high workload and time pressure across interconnected features made it hard to fix usability issues. We often had to refactor parts of our system as our system grew, debugging distributed issues with authentication, database and APIs was time consuming. In spite of all these challenges, we learned a valuable lesson about system design and team coordination.

6.0 Closing

6.1 Lessons Learned

One of the biggest things we learned during this project was the importance of planning for system complexity early on, because what started as a well thought out concept quickly grew into a multilayered application with so many interconnected components. We also underestimated the number of additional components we needed to make features like boss battles fully functional. This taught us to think past the current flow and also consider edge cases when implementing complex logic.

6.2 Conclusion

This project was not just about trying new technologies but also making these technologies work to fit our purpose. From the frontend implementation to the AWS infrastructure, we've learned to adapt tools we worked with to meet classroom requirements. While ClassQuest still has more room for improvement throughout our development cycles we've built an engaging platform that has potential in creating a gamified learning environment.

We've evolved our idea into a full-stack application with quests, boss battles, guilds and a fully working progression system, where each component supports student motivation and the teacher's workflow. We also faced some challenges with implementing certain ideas and learned to balance ambition with what is possible.

While we were able to finish what we had planned for our MVP, looking forward we still have additional game modes to be added to the boss battles, guilds systems will need to support collaboration, we will need to add character specific abilities based on roles and improve in game shop to contain cosmetic items while

also adding side quests. Overall, ClassQuest is no longer just a prototype, it's a solid foundation that we are proud of and grateful to develop.